Immigration Flows and Rental Housing Costs: Evidence from U.S. Metropolitan Areas

ECON 490 Economics Capstone Final Paper

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1. Introduction

From 2010-2019, the United States received the highest number of net migrants in its history. As has been the case throughout American history, this sustained amount of immigration caused concerns regarding the impact of migration on the wages and net welfare of US-born workers and on local housing costs. Most economic research has focused on the former rather than the latter. While some research has examined immigration's effects on local housing markets at the city level, relatively little work has studied this relationship systematically across the full range of U.S. metropolitan areas using recent data. Considering that immigrants are much more likely to be concentrated in urban areas and are more likely to rent than U.S.-born residents, it should be expected that an increase in immigration would result in an increase in housing and rental costs in those areas (Brown et al., 2014). By economic principles, an increase in immigration-related housing demand and an upward-sloping housing supply should result in increased housing costs in the short-run. However, is this relationship substantiated in real world data? Does the rate of foreign-born immigration into a metropolitan area predict faster real rent growth in the years that follow, and does this relationship vary across U.S. regions?

In order to answer this question, I estimate an OLS regression of real rent growth on the net immigration flow rate into metropolitan statistical areas (MSAs) from 2010-2014 and the percentage change in median real rental costs from 2015-2019, using data from the American Community Survey (ACS). Factoring in a number of controls, including median household income, domestic migration rate, unemployment, and the median gross rent, I find that every percentage point increase in net immigration from 2010-2014 is correlated with a 1.61 percentage point increase in median rent growth from 2015-2019 in the same MSA. However, this coefficient is insignificant at the 10 percent level.

I then use a number a geographic fixed effects specifications to account for geographic confounders across different geographic divides. While the coefficient for the immigration flow rate varies across specifications, it never falls below 3.06 and remains statistically significant at the 1 percent level in all three models. I then construct two models with interaction terms tracking how the immigrant-rent relationship varies by Census region and division, respectively. From these models, I find that the immigration–rent relationship is strongest in the South region and, at the division level, is strongest in South Atlantic metros. Finally, I discuss the design limitations of this model and the data it uses, and the supply-side complications this model fails to accurately include.

The remainder of this paper proceeds as follows. Section 2 provides background on the existing literature examining immigration's effects on local housing markets and the role of housing supply heterogeneity in moderating demand shocks. Section 3 describes the data source, variable construction, and empirical strategy, including the OLS regression framework and geographic fixed effect specifications used to identify the immigration–rent relationship. Section 4 presents the results from the baseline cross-sectional

regressions, the residuals analysis, and the regional interaction models. Section 5 concludes with a discussion of the findings, limitations of the model, and directions for future research.

2. Background

While immigration is a heavily researched topic within Economics, most research has focused on the impact of immigration on the labor market and the wages of U.S.-born residents. However, the papers which examine the impact immigration has on housing costs agree that there is a positive correlation, though they disagree on the overall economic impact on native-born Americans.

Saiz (2003) examines the impact of the Mariel boatlift, an event in 1980 where the city of Miami received approximately 140,000 Cuban refugees within the span of one year, on local rents. This made Miami a favorable natural experiment as the immigration surge which occurred in Miami was unique to the city. The paper found a causal impact where rents in Miami increased by 8%-11% from 1979-1981 more than in comparable cities; the paper also found that high-quality rental units were unimpacted by the immigration shock, while units occupied by low-income Hispanic residents faced additional rent increase of approximately 8% relative to other low-income units (Saiz, 2003).

Saiz (2007) explores the immigration-rent relationship on a national level. The paper examines the causal impact of immigration on rents within 316 metropolitan statistical areas using Census data from 1970-2000. To address the concern that immigrants be biased towards economically prospering cities or cities with an already large immigrant population, the paper uses a shift-share instrument which allocates national immigration flows to metros based on historical settlement patterns. The paper found that 1% immigration inflow is associated with approximately 1% increases in rents and housing values (Saiz, 2007).

Card (2010) uses Decennial Census data from 1980-2000 to examine the relationship between immigration inflows into Core Based Statistical Areas and the relative changes to wages, housing costs, and demographic makeup of the specific area. Building upon previous work in the field, the paper argues that the associated increase in housing costs caused by an increase in immigration is offset by a productivity increase caused by those same immigrants, evidenced by a relative increase in median wages. Card finds a 10-percentage point increase in the fraction of immigrants was associated with 4% higher native wages for the same metropolitan area in the year 2000 (2010, table 8). However, Card also finds that this average wage gain is not evenly distributed as in high-immigration cities such as Miami and Los Angeles, low-skilled workers face a 3–4% decrease in relative wages compared to low-skilled workers in low-immigration cities such as Philadelphia and Detroit (Card, 2010).

The body of mentioned literature seems to acknowledge that increased immigrant inflows are associated with, if not causing, increases in the destination area's rents. However, this consensus becomes more nuanced when considering Saiz (2003) finding that rent increases are concentrated in lower-quality

units rather than the broader rental market, and Card's documentation of uneven wage effects across skill groups. These findings suggest that the net welfare impact of immigration on native-born renters is not evenly distributed but is instead redistributive: high-skill workers experience wage gains which exceed any rent increases they face, while simultaneously low-skill workers bear higher rents in the units they occupy and lower wages in the labor markets where they compete with immigrants.

My paper seeks to build on this literature in two ways. First, I examine more recent data from 2010 to 2019, a period characterized by historically high immigration inflows and rising housing costs that the existing literature does not cover. Second, I test whether the immigration–rent relationship varies systematically across U.S. Census regions and divisions motivated by the theoretical expectation that housing supply conditions, which differ substantially across U.S. regions, should condition how demand shocks from immigration translate into rent increases, with supply-elastic markets potentially better positioned to absorb demand pressure.

3. Methods

3.1 Data

Table One: Summary Statistics

Variable	**Mean**	**Std. Dev.**	**Min**	**Max**	**N**
Outcome Variable					
Real rent growth	0.0491	0.0543	-0.0848	0.280	332
Key Explanatory Variable					
Immigration flow rate	0.0029	0.0028	0.0000	0.0223	332
Control Variables					
Domestic migration rate	0.0242	0.0147	0.0036	0.100	332
Rental vacancy rate	0.0741	0.0351	0.0165	0.312	332
Population growth	0.0401	0.0689	-0.205	0.388	332
Baseline median rent ($000s)	0.903	0.186	0.606	1.760	332
Median HH income ($000s)	54.5	9.57	36.1	100.0	332
Unemployment rate	0.0920	0.0260	0.0331	0.181	332
Unemployment rate change	-0.0371	0.0168	-0.0943	0.0076	332

Note: Summary statistics computed across 332 metropolitan statistical areas. All dollar-denominated variables are CPI-adjusted to 2019 dollars. All variables are measured during the 2010-2014 ACS 5-year estimate baseline period except real rent growth and unemployment rate change, both of which capture the change between the 2010-2014 and 2015-2019 periods within the respective variable. Immigration flow rate and migration rates are expressed as proportions of MSA population. Data from the ACS five-year estimates measured from 2010-2014 and 2015-2019, accessed through IPUMS NGHIS.

Through NHGIS IPUMS, I compile cross-sectional data of 332 metropolitan statistical areas (MSA) between two five-year ACS estimations: a baseline period between 2010-2014, and 2015-2019. I chose these two time periods as it was the most recent non-overlapping ACS 5-year estimates that are not influenced by major macroeconomic disruptions such as the Great Recession or the COVID-19 pandemic. Five-year estimates were used in order to ensure accurate estimations for MSAs below 65,000 people. Each observation represents one MSA, and there is a total of 332 observations within the data set.

Table 1 presents summary statistics for all variables used in the analysis. All variables are measured during the baseline period between 2010-2014, with the exception of real rent growth and unemployment rate change. The outcome variable, real rent growth, is constructed as the inflation-adjusted percentage change in median gross rent between the 2010–2014 and 2015–2019 ACS 5-year estimate periods. The 2010–2014 baseline rent is converted to 2019 dollars using the BLS CPI-U adjustment factor prior to calculating the percentage change to account for general inflation. The key explanatory variable is the immigration flow rate: the number of foreign-born residents who moved to the MSA from abroad expressed as a share of total MSA population, measured during 2010–2014. Measuring immigration flows during the baseline period rather than concurrently with rent growth ensures that the key regressor predates the outcome being explained, partially addressing reverse causality concerns.

Control variables include the domestic migration rate, which captures the share of native-born residents who moved from a different state. This variable is meant to isolate the impact on rent growth from foreign immigration from the impact due to domestic movement. The rental vacancy rate is constructed as vacant-for-rent units as a share of total rental housing stock, serving as a proxy for short-run housing supply availability. Median household income and baseline median gross rent are both CPI-adjusted to 2019 dollars and expressed in thousands, controlling for pre-existing income levels and rent levels respectively. The unemployment rate captures baseline labor market conditions, while the unemployment rate change – constructed as the 2019 unemployment rate minus the 2014 baseline rate – captures within-period labor market changes. As shown in Table 1, the mean unemployment rate change of -3.71 percentage points confirms that on average U.S. metros experienced improving labor markets during the 2014–2019 outcome period, consistent with a broad economic expansion during this period.

3.2 Analysis Strategy

I estimate an OLS regression of real rent growth, the inflation-adjusted percentage change in median gross rent between the two time periods, on the immigration flow rate into 332 U.S. metropolitan statistical areas during 2010–2014, controlling for a set of demographic and economic baseline characteristics described in Section 3.1.

1) $$\boldsymbol{Real\ Rent\ Growth}_m = \beta_0 + \beta_1(\boldsymbol{Immigration\ Flow\ Rate}_m) + \beta_2(\boldsymbol{Domestic\ Migration\ Rate}_m) + \beta_3(\boldsymbol{Rental\ Vacancy\ Rate}_m) + \beta_4(\boldsymbol{Population\ Growth}_m) + \beta_5(\boldsymbol{Baseline\ Median\ Rent}_m) + \beta_6(\boldsymbol{Median\ HH\ Income}_m) + \beta_7(\boldsymbol{Unemployment\ Rate}_m) + \beta_8(\boldsymbol{Unemployment\ Rate\ Change}_m) + \beta_9(\boldsymbol{Geographic\ Fixed\ Effects}_m) + \varepsilon_m$$

2) $$\boldsymbol{Real\ Rent\ Growth}_m = \beta_0 + \beta_1(\boldsymbol{Immigration\ Flow\ Rate}_m) + \beta_2(\boldsymbol{South}_m) + \beta_3(\boldsymbol{Northeast}_m) + \beta_4(\boldsymbol{West}_m) + \beta_5(\boldsymbol{Immigration\ Flow\ Rate}_m \times \boldsymbol{South}_m) + \beta_6(\boldsymbol{Immigration\ Flow\ Rate}_m \times \boldsymbol{Northeast}_m) + \beta_7(\boldsymbol{Immigration\ Flow\ Rate}_m \times \boldsymbol{West}_m) + \beta_8(\boldsymbol{Domestic\ Migration\ Rate}_m) + \beta_9(\boldsymbol{Rental\ Vacancy\ Rate}_m) + \beta_{10}(\boldsymbol{Population\ Growth}_m) + \beta_{11}(\boldsymbol{Baseline\ Median\ Rent}_m) + \beta_{12}(\boldsymbol{Median\ Household\ Income}_m) + \beta_{13}(\boldsymbol{Unemployment\ Rate}_m) + \beta_{14}(\boldsymbol{Unemployment\ Rate\ Change}_m) + \varepsilon_m$$

Equation 1 presents the baseline regression specification estimated across the 332 MSAs in the sample. The coefficient of interest is β_1, which captures the partial correlation between the immigration flow rate and real rent growth holding all other variables constant. The regression includes a set of controls intended to account for pre-existing economic conditions and to partially capture within-period economic changes. Geographic fixed effects are included to account for geographic confounding across the United States. Three specifications are estimated – Census region, Census division, and state fixed effects – to assess whether the immigration coefficient is stable as progressively finer geographic controls are introduced, with stability across specifications indicating against geographic confounding driving the result.

Equation 1 constrains the immigration-rent slope to be identical across all U.S. regions, which may obscure geographic heterogeneity within the relationship. Equation 2 extends the baseline specification by interacting the immigration flow rate with Census region indicators, allowing β_1 to vary across the four U.S. Census regions, motivated by the theoretical expectation that differing housing supply conditions should modulate how immigration-driven demand shocks translate into rent increases. $South_m$, $Northeast_m$, and $West_m$ are indicator variables equal to one if MSA m falls in that Census region and zero otherwise, with Midwest as the omitted reference category. The coefficients β_2 through β_4 capture baseline rent growth differences across regions independent of immigration flows, while β_5 through β_7 capture how much steeper

or flatter the immigration–rent slope is in each region relative to the Midwest baseline. The total implied immigration-rent slope for any region is therefore β_1 plus the respective interaction coefficient. For example, the total slope for Southern metros is $\beta_1 + \beta_5$. If β_5 is positive and statistically significant, this indicates that the immigration–rent relationship is stronger in Southern metros than in the Midwest baseline. An analogous specification replacing Census region interactions with Census division interactions is also estimated as a robustness check, decomposing the regional findings into nine constituent divisions.

4. Results

3.1 Single Slope Estimate Models

Table 2: Single Slope Estimate Models

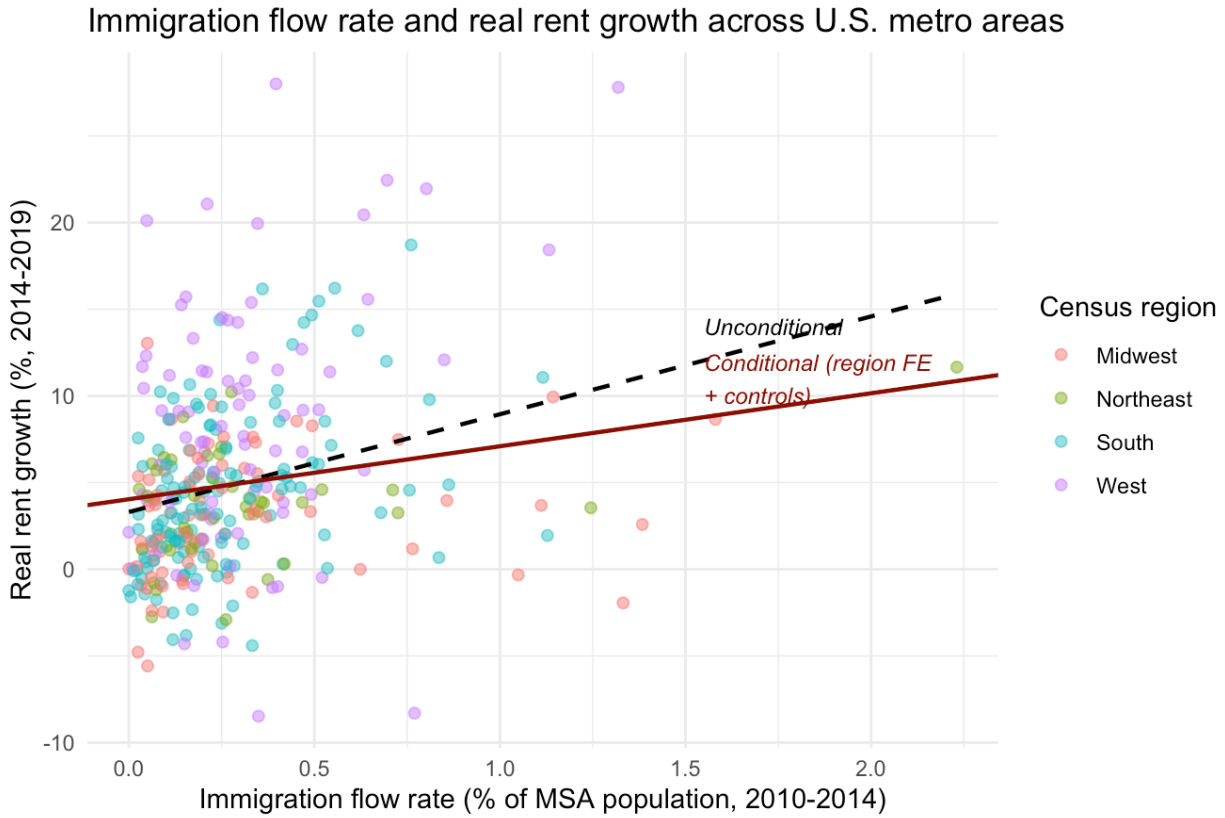
Variable	**Model 1 No FE**	**Model 2 Region FE**	**Model 3 Division FE**	**Model 4 State FE**
Key Explanatory Variable				
Immigration flow rate	1.621 (1.166)	3.276*** (1.040)	3.249*** (1.017)	3.007*** (0.967)
Controls				
Domestic migration rate	-0.147 (0.212)	-0.327* (0.190)	-0.212 (0.210)	-0.152 (0.215)
Rental vacancy rate	-0.174** (0.069)	-0.119* (0.071)	-0.101 (0.074)	-0.039 (0.081)
Population growth	0.107** (0.042)	0.053 (0.036)	0.054 (0.036)	-0.023 (0.033)
Baseline median rent (real, $000s)	0.097*** (0.022)	0.007 (0.026)	-0.013 (0.029)	-0.041 (0.032)
Median HH income (real, $000s)	0.001 (0.001)	0.002*** (0.001)	0.002*** (0.001)	0.003*** (0.001)
Baseline Unemployment rate	-0.895*** (0.189)	-1.040*** (0.185)	-1.053*** (0.189)	-0.797*** (0.220)
Unemployment rate change	-1.103*** (0.249)	-1.283*** (0.249)	-1.390*** (0.249)	-0.803*** (0.265)
Model Fit				
Observations	332	332	332	332
R-squared	0.351	0.455	0.468	0.651
Adjusted R-squared	0.335	0.436	0.442	0.574
Mean (Median Rent Growth)	4.91%	4.91%	4.91%	4.91%

Note: HC1 heteroskedasticity-robust standard errors in parentheses. All dollar controls CPI-adjusted to 2019 dollars and expressed in thousands. Reference categories: Region FE = Midwest; Division FE = East North Central; State FE = Alaska. *** p<0.01 ** p<0.05 * p<0.10

Table 2 captures the result of the single slope estimate models with median real rent growth as the outcome variable. The main coefficient of interest is for immigration flow rate. A one percentage point increase in an MSAs immigration flow rate would be correlated with the coefficient shown. One important

finding is the change in the immigration flow rate coefficient when geographic fixed effects were applied. In model one, the coefficient for immigration flow rate is 1.621 and is not statistically significant at any conventional threshold. However, the coefficient increases to 3.276 with region fixed effects while also becoming statistically significant at a 1% level. The coefficient remains significant at a 1% level while never dropping below 3.007 after applying more granular geographic fixed effects. This change suggests the cross-regional differences in baseline rent growth and immigration levels are sufficiently correlated with other geographic factors to the extent where the no fixed effects model cannot sufficiently isolate the immigration–rent correlation. This would mean the immigration-rent relationship changes between specific geographic regions rather than being uniform across the United States. These differences between geographic regions will be further explored with the geographic interaction terms.

Figure One



Notes: The solid black regression line represents the fitted values from a simple bivariate OLS regression of real rent growth on the immigration flow rate without controls or fixed effects. The solid red line represents the fitted values from an OLS regression with controls and region fixed effects. Data sourced from IPUMS NHGIS ACS 5-year estimates.

Figure 2 illustrates the immigration-rent relationship across the 332 MSAs in the sample. The black dashed line represents the unconditional OLS regression line from a bivariate regression without controls or fixed effects, while the dark red solid line represents the conditional partial correlation from the region

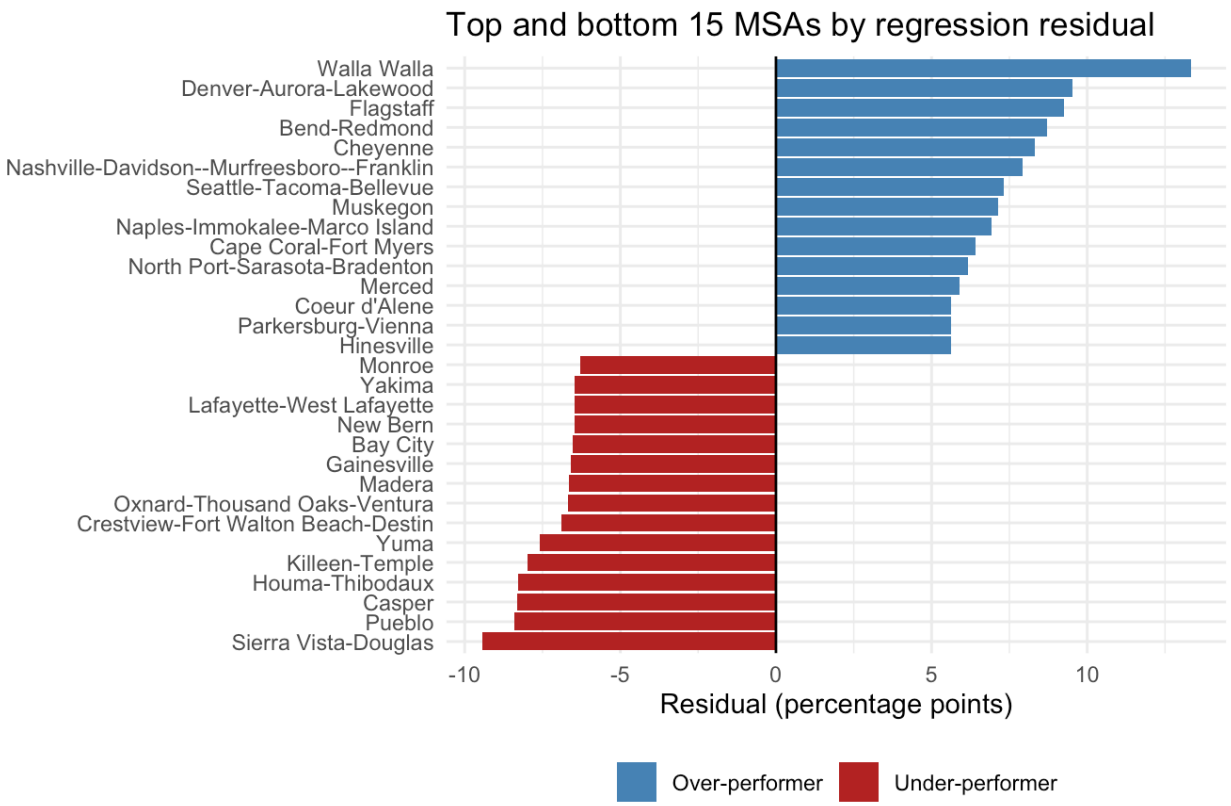
fixed effects model. The gap between the two lines reflects the geographic confounding absorbed by the fixed effects. The unconditional slope is steeper because it conflates the within-region immigration-rent relationship with cross-regional baseline differences in both variables.

Additionally, several control variables produce results consistent with economic intuition and remain stable across specifications. The baseline unemployment rate is negative and statistically significant across all four models, indicating that metros with weaker labor markets at baseline experienced slower real rent growth, supporting the theory that stronger labor markets may drive increased demand for housing. The within-period unemployment rate change is similarly negative and significant throughout, suggesting that metros whose labor markets deteriorated during 2014–2019 experienced slower rent growth independent of where they started. This finding indicates that MSAs which experienced slowing labor markets, and by extension weaker economic conditions, have weaker housing markets. Median household income is positive and significant in the region, division, and state fixed effect specifications, consistent with the expectation that higher-income metros experience faster rent growth as residents can afford to bid more for housing.

Two variables lose significance as geographic controls become more granular. The rental vacancy rate is significant at the 10% level under region fixed effects but loses significance under state fixed effects, suggesting that vacancy rate variation is partly captured by state-level housing market conditions absorbed by the finer fixed effects. Population growth is similarly significant without fixed effects but loses significance once geographic controls are introduced, consistent with population growth being correlated with regional economic conditions that the fixed effects absorb. Domestic migration rate and baseline median rent are not statistically significant in any specification, suggesting that after controlling for income, unemployment, and geography, these variables add limited independent explanatory power for rent growth across this sample.

3.2 Residual Analysis

Graph 2: Over and Under Performers Bar Graph



Note: Figure above displays the top 15 over-performing and bottom 15 under-performing metropolitan statistical areas (MSAs) ranked by their regression residual from the state fixed effects model. Each bar represents one MSA. Blue bars extending to the right indicate MSAs whose actual real rent growth fell short of the model's predicted value. Red bars extending to the left indicate MSAs whose actual real rent growth fell short of the model's predicted value. Residuals are expressed in percentage points. Data sourced from IPUMS NHGIS ACS 5-year estimates.

Prior to estimating the interaction models, I examine the residuals from the state fixed effects model (Model 4, Table 2) to identify which metropolitan areas systematically over- or under-perform relative to the model's predictions. Over-performers are MSAs where actual real rent growth exceeded the model's fitted value given their modeled characteristics. Under-performers are MSAs where actual rent growth fell short of the fitted value. Figure 4 displays the top and bottom 15 MSAs ranked by residual magnitude.

The residuals reveal clustering around within-period economic shifts. Over-performing metros tend to cluster around three broad patterns. First, technology sector and lifestyle migration booms in Colorado and the Pacific Northwest: Denver, Flagstaff, Cheyenne, Bend-Redmond, and Seattle all appear in the top 15 . This could be explained by an increase in domestic in-migration between periods from high-cost coastal cities driving rent pressure beyond what the immigration flow rate and baseline economic conditions predict.

Second, retirement-driven domestic in-migration along the Southwest Florida Gulf Coast — Naples, Cape Coral-Fort Myers, and North Port-Sarasota-Bradenton all over-perform, reflecting seasonal and permanent retiree demand that the ACS permanent population measures do not fully capture (Hayward, Martínez, and Spell, 2026). Third, amenity-driven Pacific Northwest spillover markets such as Walla Walla, Washington, where proximity to larger high-cost metros may generate rent pressure disproportionate to the metro's size and immigration flows.

Under-performing metros cluster around negative economic shocks. The 2014–2016 collapse in global oil prices is the most visible pattern; Casper, Wyoming and Houma-Thibodaux, Louisiana both experienced dramatic economic deterioration during the outcome period as energy sector employment contracted, theoretically suppressing housing demand below what their observed baseline characteristics predicted (Murphy, Plante, and Yücel, 2015). Military base population fluctuations could drive underperformance in Sierra Vista-Douglas, Arizona and Killeen-Temple, Texas, where federal deployment decisions play an outsized role in aggregate housing demand relative to most metros (Military OneSource, 2024). The impact of these bases would lead to median rent fluctuations that is largely uncorrelated with the demographic and economic controls in the model, and therefore appears as unexplained negative residual variation when actual rent growth falls short of what those controls would predict.

The systematic nature of this clustering has two important implications for interpreting the regression results. First, it confirms that the model is omitting within period changes to the MSAs baseline characteristics. The inclusion of the unemployment rate change variable likely partially addresses this limitation but does not fully resolve it. Second, the economic conditions explanation raises the possibility of upward bias in the immigration flow rate coefficient. Economically dynamic metros that attract immigrants may also over-perform on rent growth for reasons unrelated to immigration itself. For example, an area experiencing mass economic growth and higher median wages would find its residents able to bid more for housing, thus raising housing costs independent of increased immigration. To the extent that these economic dynamics are correlated with immigration flows but not fully absorbed by the unemployment change control, the β_1 coefficient may partially capture rent increases driven by broader economic expansion rather than immigration-specific demand pressure.

It is important to note that the economic stories presented for these residuals are plausible interpretations consistent with the data rather than established conclusions. Confirming these mechanisms would require additional data on within-period employment growth by sector, domestic migration by age group, and energy sector exposure, none of which are available within the current dataset. This residuals analysis is therefore best understood as a qualitative exercise to explore the limitations of this model rather than any concrete explanation of unexplained variation.

3.3 Region and Division Interaction Models

Table 2: Region Interaction Model

Variable	Estimate
Key Regressor	
Immigration flow rate (Midwest baseline)	0.746
	-1.318
Immigration flow rate × Northeast	1.879
	-1.771
Immigration flow rate × South	5.038**
	-2.234
Immigration flow rate × West	5.059
	-3.245
Controls	
Median HH income (real, $000s)	0.002***
	-0.001
Unemployment rate	-1.036***
	-0.187
Unemployment rate change	-1.269***
	-0.247
Region Fixed Effects (Main Effects)	
Northeast	-0.005
	(0.008)
South	0.018**
	(0.008)
West	0.041***
	(0.012)
Model Fit	
Census Region fixed effects	Yes
Additional controls	Yes
Observations	332
R-squared	0.467
Adjusted R-squared	0.443
Mean (outcome)	4.91%

Note: This table presents results from an OLS regression estimated on a cross-section of 332 U.S. metropolitan statistical areas (MSAs) that extends the baseline model (Table 2) by allowing the immigration flow rate coefficient to vary across the four U.S. Census regions through interaction terms. All dollar controls are CPI-adjusted to 2019 dollars and expressed in thousands. The baseline region is the Midwest. Standard errors in parentheses are HC1 heteroskedasticity-robust. *** p<0.01 ** p<0.05 * p<0.10

Table 3 presents the region interaction model results. The Midwest serves as the reference category with a baseline immigration–rent slope of 0.746, which is not statistically significant. A notable finding is

the South region interaction coefficient of 5.038, statistically significant at the 5% level (p = 0.025). The implied total immigration–rent slope for Southern metros is approximately 5.8 percentage points, meaning a one percentage point increase in the immigration flow rate is associated with approximately 5.8 percentage points more real rent growth in Southern metros, holding all controls constant. This is substantially above the national average of approximately 3.3 percentage points from Stage 1a and more than seven times the Midwest baseline slope. The South region main effect is also positive and significant (0.018*, p = 0.028), indicating that Southern metros experienced faster baseline rent growth than Midwestern metros independent of immigration flows during this period.

The West interaction coefficient of 5.059 is similar in magnitude to the South but does not reach conventional significance thresholds (p = 0.120), with a confidence interval spanning from approximately -1.3 to +11.4. The wide confidence interval likely reflects substantial heterogeneity within the West region: supply-constrained Pacific coast metros in California, Oregon, and Washington may show a strong immigration–rent relationship while theoretically more supply-elastic Mountain metros in Colorado, Arizona, and Nevada may show a more moderate response (U.S Census Bureau, 2024). The West main effect is positive and highly significant (0.041**, p < 0.001), confirming that Western metros experienced substantially faster baseline rent growth than Midwestern metros during this period regardless of immigration flows. The Northeast interaction is small and insignificant (1.879, p = 0.290), suggesting no detectable difference in the immigration–rent relationship between Northeastern and Midwestern metros.

Table 3: Census Division Interaction Model

Variable	**Estimate**
Key Regressor	
Immigration flow rate (East North Central baseline)	1.693
	(1.707)
Immigration flow rate × East South Central	6.626
	(5.296)
Immigration flow rate × Middle Atlantic	1.475
	(1.996)
Immigration flow rate × Mountain	0.403
	(7.453)
Immigration flow rate × New England	-4.740
	(4.087)
Immigration flow rate × Pacific	5.142
	(3.130)
Immigration flow rate × South Atlantic	5.920*
	(3.038)
Immigration flow rate × West North Central	-2.654
	(2.682)

Immigration flow rate × West South Central	1.466
	(3.274)
Controls	
Median HH income (real, $000s)	0.002***
	(0.001)
Unemployment rate	-0.988***
	(0.195)
Unemployment rate change	-1.338***
	(0.250)
Model Fit	
Census division fixed effects	Yes
Additional controls	Yes
Observations	332
R-squared	0.485
Adjusted R-squared	0.445
Mean (outcome)	4.91%

Note: This table presents results from an OLS regression estimated on a cross-section of 332 U.S. metropolitan statistical areas (MSAs) that extends the baseline model (Table 2) by allowing the immigration flow rate coefficient to vary across the nine U.S. Census divisions through interaction terms. All dollar controls are CPI-adjusted to 2019 dollars and expressed in thousands. The baseline region is the Midwest. Standard errors in parentheses are HC1 heteroskedasticity-robust. *** p<0.01 ** p<0.05 * p<0.10

Table 4 presents the division interaction model results. At the division level, the South Atlantic interaction produces the largest implied slope at approximately 7.6 percentage points, suggesting the South region finding from Stage 2b is concentrated in Atlantic coast metros including Florida, Georgia, North Carolina, and Virginia. However, this interaction term falls just short of conventional significance thresholds ($p = 0.052$) and should be interpreted as suggestive rather than a confirmed finding. Notably, the West South Central division - containing the fast growing and supply-elastic Texas metros of Houston and Austin - shows an implied slope of approximately 3.2, close to the national average and substantially below the South Atlantic (U.S Census Bureau, 2024). This finding may suggest that more permissive development environments can moderate the short-run immigration–rent relationship.

The remaining division interaction terms are not statistically significant, with wide confidence intervals reflecting thin cell counts within individual divisions rather than precisely estimated null effects. For example, the Mountain division confidence interval spans from approximately -14 to +15, encompassing a varying range of plausible slopes. These results illustrate why the region-level interaction model in Stage 2b remains the primary specification: individual divisions contain too few MSAs to reliably estimate division-specific slopes. The division model is best understood as exploratory evidence consistent with the South region finding rather than proving anything definitive by itself.

5. Conclusion

This paper examines whether U.S. metropolitan areas that received higher rates of foreign-born immigration during 2010–2014 experienced faster real rental price growth between 2014 and 2019. Using OLS regression across 332 metropolitan statistical areas with data from the American Community Survey, I find evidence consistent with a positive immigration–rent relationship that is robust across geographic specifications and concentrated in specific regions of the country.

The baseline results indicate that a one percentage point increase in the immigration flow rate is associated with approximately 3.1 to 3.3 percentage points more real rent growth, statistically significant at the 1% level across region, division, and state fixed effect specifications. The modest decline in the coefficient as geographic controls become more granular suggests a small degree of geographic confounding, though the stability of the estimate across specifications suggests a robust partial correlation. Regional interaction results indicate this relationship is concentrated in Southern metros, where the implied immigration–rent slope reaches approximately 5.8 percentage points, more than double the national average, with division-level results providing evidence that the South effect is concentrated in South Atlantic metros, although this division-level finding falls short of conventional significance thresholds.

Several limitations should be acknowledged when interpreting these results. The cross-sectional design cannot establish causality. The enclave effect documented by Card, whereby immigrants systematically select destinations with existing immigrant communities, represents a potential source of upward bias that instrumental variable methods such as those employed by Saiz (2007) would be needed to address. The model also struggles to capture within-period economic changes beyond the unemployment rate change control included here; accelerating domestic migration, income growth, and sector-specific economic shocks during the outcome window may all be correlated with immigration flows in ways that inflate the estimated coefficient. Finally, the five-year study window captures primarily the short-run demand effect of immigration on rents before housing supply has had meaningful time to respond. Supply-elastic markets such as Houston and Austin may absorb immigration-driven demand through new construction over longer horizons, potentially leading to smaller the regional slopes than the ones documented in this project.

Taken together, the evidence is consistent with positive correlation between international immigration flows and rental housing costs in U.S. metropolitan areas during this period, with the relationship strongest in Southern and coastal metros where housing supply constraints may amplify the rent response to population-driven demand shocks. Whether this rent increase represents a net welfare loss for native-born renters depends on the distributional wage effects documented by Card. High-skill workers who experience wage gains from immigration may find rent increases manageable, while low-skill workers

who face both wage compression and higher rents in lower-quality units may experience a genuine deterioration in housing affordability. Resolving this distributional question and disentangling the causal immigration effect from the broader economic conditions of high-immigration metros, remains an important direction for future research.

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